BUI THANH HAI

Mobile developer

0865831896 buithanhhai2172002@gmail.com https://github.com/haivc2002 HaNoi, VietNam

SUMMARY

Flutter Developer with 3 years of experience building and maintaining production mobile applications. Strong in Flutter/Dart, BLoC/Riverpod, Clean Architecture, and feature-first architecture, with hands-on experience in real-time video, multimedia processing, and native platform integration across Android and iOS.

EDUCATION

Electric Power University 2020 - 2024

INFORMATION TECHNOLOGY

Major in Software Engineering
Toeic Academic - 600

EXPERIENCE

ATIN – Advanced Technology Innovations Hanoi, Vietnam 01-2025 - Now

Mobile Developer (Android & Flutter)

- Developed and maintained production **Flutter/Dart, Kotlin, and Android/iOS** applications for enterprise and government clients, including **Viettel, SunGroup**, and other partners.
- Built real-time **Video Management Systems** using **Flutter, WebRTC, RTSP, VLC/LibVLC, and WebSocket** for camera monitoring and device management.
- Developed multimedia and AI-powered audio applications using **Flutter, FFmpeg**, audio processing, and on-device media processing.
- Designed and implemented scalable **modular and feature-first architectures** with **BLoC/Riverpod, SQLite, local storage,** and code generation across multiple production applications.
- Developed native Android and iOS integrations, bridging Flutter with platform capabilities and implementing system-level mobile features.
- Developed native Android applications with **Kotlin**, applying **Coroutines, Jetpack, ViewModel, Hilt, MVVM, and Clean Architecture**, with integrations for Camera, NFC, and face recognition SDKs.

XTEL Software Solutions 11-2023 - 12-2024

Mobile developer

- Developed and maintained production **Flutter/Dart** applications for government organizations and outsourcing clients.
- Built cross-platform mobile applications using **Flutter**, integrating **RESTful APIs, Firebase, and third-party services**.
- Implemented scalable mobile architectures using **BLoC/GetX, Clean Architecture, and feature-based architecture**.
- Developed core mobile functionalities including authentication, networking, push notifications, local storage, and application infrastructure.
- Diagnosed and resolved production issues, performed performance optimization, and delivered continuous feature development and maintenance.
- Collaborated with **Backend and UI/UX teams** to integrate APIs, implement product requirements, and deliver production-ready mobile applications.

PROJECTS

Smart Guard — Real-time Video Monitoring & Security Platform | Flutter, Dart, WebRTC, BLoC, Dio

- Built a high-performance cross-platform security monitoring application with **Flutter/Dart**, **Modular Architecture**, and **BLoC**, supporting real-time camera monitoring and device management.

- Integrated **WebRTC** for real-time video streaming and engineered custom stream lifecycle management across **Foreground/Background** application states.
- Optimized video rendering and resource consumption by dynamically controlling WebRTC tracks and combining **App Lifecycle, TickerMode, and ModalRoute** to pause inactive streams and minimize unnecessary **CPU/GPU and battery usage**.
- Implemented robust networking with **Dio**, custom **interceptors**, automatic token refresh, request queuing, and retry handling for authenticated API communication.
- Built native platform integrations through **MethodChannel**, including biometric authentication and platform-adaptive UI, with secure credential storage using **Keychain/Keystore**.

AI Record — AI-powered Audio Recording & Media Processing Application | Flutter, Dart, BLoC, FFmpeg, GLSL, SQLite

- Built a production **Flutter/Dart** application for smart audio recording and media management, using **BLoC** and **Modular Architecture** to organize reusable application components.
- Integrated **FFmpeg Kit** for on-device audio processing, format conversion, and media manipulation, reducing dependency on server-side processing for media operations.
- Developed custom **Fragment Shaders (GLSL)** for GPU-accelerated audio visualizations and dynamic visual effects, including **Silk Wave** and **Aura Lumina** rendering.
- Designed an **offline-first** data and file management layer using **SQLite**, handling recording metadata, local audio files, playback, and persistent application data.
- Implemented background audio capabilities, system volume control, runtime permissions, and cross-platform file sharing to provide a complete mobile recording experience.

Device Management System Android OTA Agent & Monitoring Platform (Built for SunGroup's SunWorld) | Kotlin, JavaScript, Node.js, MongoDB

Team size: 2

- Developed an **Android device management agent** using **Kotlin and Coroutines** for background execution, supporting silent APK installation, scheduled tasks, and automated **OTA updates** on Android-based face terminals.
- Built persistent **background services** to collect device telemetry including CPU, RAM, and storage usage, stream real-time device status via **WebSocket**, and collect device logs.
- Integrated the agent with a web-based monitoring platform built with **Node.js**, enabling remote OTA deployment, device health monitoring, and online/offline status management across multiple deployments.
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PACKAGES

fog_edge_blur (Flutter Plugin) | Flutter - Dart, Kotlin, GLSL

Pub.dev: https://pub.dev/packages/fog_edge_blur

Cross-platform edge blur plugin using **GLSL shaders** and native **Kotlin integration**, optimized for smooth rendering.

flow_box_popup (Flutter Plugin) | Flutter - Dart

Pub.dev: https://pub.dev/packages/flow_box_popup

Cross-platform popup animation plugin for **iOS and Android**, featuring a simple API and smooth rendering.

fluid_wave (Flutter Plugin) | Flutter - Dart, GLSL

Pub.dev: https://pub.dev/packages/fluid_wave

Custom fluid wave and UI effects plugin using **GLSL-based rendering**, optimized for smooth animations.

View project details: https://haivc2002.github.io/CV/?tab=projects

TECHNICAL SKILLS

Mobile Development	Flutter, Dart, Kotlin, Riverpod, BLoC, GetX; Jetpack (ViewModel, Navigation, Room, DataStore, WorkManager); MVVM, Clean Architecture; Coroutines, Flow
Networking & Data	REST APIs, JSON, WebSocket, Dio, Retrofit, OkHttp; Local Storage, SQLite
Services & Tools	Git, Firebase (Analytics, Crashlytics, Remote Config, Cloud Messaging); Git, Gradle, Android Studio